

Module 1: HTML Basics

1. Introduction to HTML

HyperText Markup Language (HTML) is the standard markup language used to create the structure and content of web pages. It acts as the skeleton of a website, defining where text, images, and other elements should appear.

2. Structure of an HTML Document

Every HTML file follows a standard boilerplate structure. This ensures browsers interpret the code correctly.

- `<!DOCTYPE html>`: Tells the browser that the document is an HTML5 file.
- `<html>`: The root element that wraps all content.
- `<head>`: Contains metadata (data about the data), such as titles and character sets.
- `<body>`: Contains the visible content of the web page.

3. Common Tags

Tags are the building blocks of HTML. They usually come in pairs: an opening tag `<tag>` and a closing tag `</tag>`.

- `<html>`: Defines the start and end of the HTML document.
- `<head>`: Contains machine-readable information (metadata).
- `<title>`: Sets the name of the page shown on the browser tab.
- `<body>`: Encloses all the content shown to the user.
- `<h1>` to `<h6>`: Used for headings. `<h1>` is the most important (largest), and `<h6>` is the least important (smallest).
- `<p>`: Defines a paragraph of text.
- `<br>`: A "break" tag used for a single line return. It does not have a closing tag.
- `<hr>`: A "horizontal rule" tag used to separate content with a visual line.

4. Lists

Lists are used to group related items together.

Ordered List ()

Used for items that follow a specific sequence (1, 2, 3...).

- Example: Steps in a recipe.

Unordered List ()

Used for items where the order does not matter (bullet points).

- Example: A grocery shopping list.

5. Hyperlinks

Hyperlinks allow users to navigate between pages. They are created using the <a> (anchor) tag.

- Example: Click here to visit Google
- The href attribute specifies the destination URL.

6. Images

Images are embedded using the tag. This tag is self-closing and requires two main attributes:

- src: The path to the image file.
- alt: A text description of the image for accessibility.
- Example:

7. Tables

Tables are used to display data in a grid format of rows and columns.

Element	Purpose
<code><table></code>	The container for the table.
<code><tr></code>	Defines a table row.
<code><th></code>	Defines a table header (bold and centered).
<code><td></code>	Defines a table cell (standard data).

8. Forms

Forms are essential for collecting user input.

- `<form>`: Wraps the entire input area.
- `<label>`: Provides a description for an input field.
- `<input>`: The most versatile element; can be text, password, or submit buttons.
- `<textarea>`: Used for multi-line text input.
- `<select>`: Creates a dropdown list for the user to choose from.

9. Example HTML Program

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>My First Web Page</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Welcome to My Website</h1>
```

```
<p>This is a paragraph about my journey in web development.</p>
```

```
<h2>My Favorite Hobbies</h2>
```

```
<ul>
```

```
  <li>Coding</li>
```

```
  <li>Reading</li>
```

```
  <li>Traveling</li>
```

```
</ul>
```

```
<p>Visit our <a href="contact.html">Contact Page</a> for more info.</p>
```

```
</body>
```

```
</html>
```

10. Best Practices

1. **Indentation:** Always indent nested tags to make code readable.
2. **Lowercase Tags:** Write all tag names in lowercase (e.g., `<body>` instead of `<BODY>`).
3. **Use Alt Text:** Always provide `alt` descriptions for images to improve accessibility.
4. **Semantic Tags:** Use tags like `<header>`, `<main>`, and `<footer>` to give meaning to your layout.

Congratulations! You have completed Module 1.